
How Do Visual Models Use Cues, Adapt to Domain Shift, Generalize, and Reject Unknowns?

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Abstract

A model can classify images accurately while relying on visual cues or information that may fail under perturbation, domain shift, or unknown classes. We study these behaviors through four experiments on visual learning and recognition. First, controlled interventions on STL-10 evaluate color, shape, texture, spatial structure, and feature representations to examine how models perceive information and their use of it during classification. We then use PACS to perform domain adaptation and generalization where we compare adaptation with unlabeled Sketch images against generalization when Sketch is unavailable. Finally, we evaluate whether CIFAR-10 classifiers reject near and far unknown CIFAR-100 classes. We find that patch shuffling causes substantially larger accuracy losses than color changes or translation. On PACS, moderate target alignment improves Sketch accuracy, whereas stronger alignment can collapse classification. Sharpness-aware training improves unseen Sketch performance. For open-set recognition, stronger augmentation helps most with far unknowns, while near unknowns remain difficult to reject. These results show that preserving class information matters as much as learning an effective feature space or reducing sensitivity to visual or domain changes. <https://github.com/WasifAhmed1917/ATML/tree/main>

1 Introduction

Image classifiers are usually evaluated on examples drawn from the same classes and distribution as their training data. Good performance in that setting does not tell us which visual information controls their decisions, whether that information transfers to a new domain, or whether the model can recognize an input outside its known label space. These questions become important when an object’s appearance, surroundings, or illustration changes while its identity or label remains the same.

We examine four related settings. Firstly, we explore inductive biases of a model. We intervene on color, shape, texture cues, object position, and spatial arrangement to study predictions alongside feature representations. Cue-conflict images are motivated by prior work on shape and texture bias (1), and thus we explore that work on STL-10 using AdaIn stylization. We then explore domain adaptation where we ask whether unlabeled target images can help a classifier trained on labeled source domains recognize Sketch images through feature alignment (4). In domain generalization, we remove target access and instead test whether alignment among observed PACS sources or local parameter stability supports generalization to unseen Sketch images (7; 8). Finally we consider open set recognition: whether classifiers can reject images from unknown classes while retaining performance on known ones (9; 10).

Across these tasks, the central question is what information a model should preserve, what changes it should become less sensitive to, and when it should decline to assign a known label. We compare overall performance with class-level failures, since a more invariant representation or a more accurate closed-set classifier does not by itself mean it makes reliable decisions in every setting.

2 Inductive Biases and Feature Representations

In this section, we formally motivate the concept of inductive biases and how different attributes and categories of inductive biases change the way a model perceives information both in its feature representation $f(X)$ and in making a decision Y , often modeled through the conditional distribution $P(Y | X)$. Inductive biases are a set of assumptions or preferences that influence how a model selects a hypothesis function h from the hypothesis space H , the set of possible hypothesis functions used to fit the training data, with the goal of generalizing well to unseen data.

Here, we consider semantic biases and architectural biases. Semantic biases refer to a model’s preference for attributes such as the shape of an object, the texture of an image, or its background color when making a decision. Shape bias quantifies a model’s preference for an object’s shape when shape and texture suggest different classes. Similarly, texture bias quantifies its preference for the texture cue under the same conditions. Formally, they are defined as follows:

$$\text{Shape Bias}(\%) = \frac{N_{\text{shape}}}{N_{\text{shape}} + N_{\text{texture}}} \times 100, \quad \text{Coverage}(\%) = \frac{N_{\text{shape}} + N_{\text{texture}}}{N_{\text{total}}} \times 100.$$

On the other hand, we have architectural biases. These are characteristics of a model’s architecture that affect how it processes information internally. For example, let a CNN receive an image

$$X \in \mathbb{R}^{H \times W \times C_{\text{in}}}$$

and apply C_{out} kernels of spatial size $K \times K$. With stride s , padding p , and no dilation, one output value is

$$A_{i,j,o} = b_o + \sum_{c=1}^{C_{\text{in}}} \sum_{u=0}^{K-1} \sum_{v=0}^{K-1} W_{u,v,c,o} X_{is+u-p, js+v-p, c},$$

where positions outside the image are supplied by the padding rule. The output has spatial size

$$\left(\left\lfloor \frac{H + 2p - K}{s} \right\rfloor + 1 \right) \times \left(\left\lfloor \frac{W + 2p - K}{s} \right\rfloor + 1 \right).$$

Thus, one output value initially depends on a local region of the input. Its **receptive field** is the region of the original image that can influence that value. As layers are stacked, this region grows, allowing later features to combine information from more distant parts of the image. Pooling and strided layers can expand it further.

The way a model uses visual cues to make a decision can also depend on how its feature representations are learned. For example, ResNet-50 processes an image through convolutional layers, whereas ViT-B/16 uses self-attention to relate information across image patches. CLIP also uses a ViT image encoder, but its representations are learned by aligning images with text through contrastive training, rather than by predicting ImageNet classes. These differences may affect which properties of an image, such as shape, texture, color, or spatial arrangement, are retained in the representation and used for classification. This motivates four central research questions of this section:

- **RQ1:** Which visual cues (texture, shape, color) do the models use when making a classification decision?
- **RQ2:** To what extent do their predictions depend on the spatial arrangement and position of those cues?
- **RQ3:** When visual cues change, do the models’ feature representations change in ways that agree with their predictions?
- **RQ4:** Which observed differences can reasonably be attributed to architecture, and which may instead reflect pretraining data, supervision, augmentation, or model capacity?

2.1 Experimental Design and Setup

To address our research questions, we design experiments and controlled interventions. However, first we construct a clean baseline reference. We start by evaluating the three classifier heads and zero-shot CLIP on the clean evaluation subset. We report top-1 accuracy, macro-F1, and mean maximum confidence. We use these results as a reference against later controlled interventions.

To address RQ1, we perform a color intervention and construct cue-conflict images to examine the bias of a model. We alter color while preserving object geometry to observe whether chromatic information contributes to the models’ decisions. We choose grayscale as the common intervention, then choose fixed hue rotation as an additional transformation to assess sensitivity. We report the change in accuracy along with prediction consistency relative to our clean images.

To analyze how reliant a model is on shape and texture for its decisions, we generate cue-conflict images in which the data supplies the shape of one class and the style supplies the texture of another. We select STL-10 as a 10-class dataset. We use AdaIN for style transfer, select five unordered class pairs, and generate both directions when feasible: shape/content = A, texture/style = B; shape/content = B, texture/style = A. Visually accepted conflicts are balanced across class pairs and directions as closely as possible. The visual rejection rule is as follows: given class A for shape/content and class B for texture/style, a conflict is accepted if the texture is visibly transferred to class A while retaining the object’s recognition and content; otherwise, the pairing is rejected as an invalid conflict. This review was conducted before seeing any model predictions.

For model evaluation, we record the accepted and rejected counts. We classify each prediction as the content/shape label, the style/texture label, or another label. We report shape bias and coverage.

For RQ2, we test whether model predictions remain stable after an object is displaced from its original position in the image. We translate each image by 0, 8, 16, and 32 pixels in the four cardinal directions using reflection padding followed by a shifted crop, and average the results across directions. We plot accuracy and prediction consistency against displacement, where

$$\text{Consistency}(\delta) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}(x_i) = \hat{y}(T_\delta(x_i))].$$

Moreover, to measure how a model’s predictions are affected by spatial arrangement, we perturb global spatial organization while retaining the image’s pixels and much of its local patterns. We divide each image into a 4×4 pixel-space grid and generate one non-identity patch permutation per image using seed 6304. We reuse exactly the same shuffled images across models and report the accuracy drop and prediction consistency relative to clean inputs.

Lastly, for RQ3, to investigate whether changes in visual cues alter feature representations, we pair each transformed image and its clean counterpart and measure the cosine stability of each backbone’s representation:

$$I_T = \frac{1}{N} \sum_{i=1}^N \frac{f(x_i)^\top f(T(x_i))}{\|f(x_i)\|_2 \|f(T(x_i))\|_2}.$$

These transformed images are obtained via color intervention, cue conflicts through stylization, translation, and patch shuffling. Moreover, we visualize both transformed and clean representations using a t-SNE map.

We use five cue-conflict class pairs: airplane–ship, bird–cat, deer–horse, dog–monkey, and car–truck, generating both shape–texture directions for each pair. We examine whether the models favor shape or texture labels without assuming a ranking between architectures, since their pretraining and classification rules also differ. We expect patch shuffling to reduce accuracy more than grayscale conversion because it disrupts spatial relationships between patches, while performance should remain largely stable under translation. Across interventions, we also examine whether higher feature cosine similarity tends to accompany greater prediction consistency. A summary of the configuration details is provided in Table 1.

2.2 Results and Discussion

Table 2 displays our clean baseline alongside color intervention and patch-shuffling results. The models perform well in accuracy and macro-F1 on the evaluation subset. While CLIP has strong zero-shot performance (0.98), its trained linear head has low mean maximum confidence (0.215). The head’s logits have a different scale from zero-shot similarities, so low softmax confidence does not mean low accuracy.

Grayscale leads to little degradation across the models, mainly 1.0–2.2 percentage points relative to their baselines. Removing chromatic information may disturb scene details that were relevant to a

Table 1: Experimental settings.

Setting	Configuration
Dataset	STL-10, 4,000 training and 1,000 validation images
Test subset	500 official test images, 50 per class
Seed and image size	6304; shared 224×224 RGB images
Frozen backbones	ResNet-50 ImageNet1K V2, ViT-B/16 ImageNet1K V1, OpenCLIP ViT-B/32 openai
Classifier heads	Linear, AdamW, batch size 128, learning rate 10^{-3} , weight decay 10^{-4}
Training	Maximum 50 epochs, checkpoint selected by validation accuracy
Selected epochs	ResNet-50: 7, ViT-B/16: 19, CLIP head: 2
CLIP zero-shot prompt	a photo of a {class}.
Color interventions	Grayscale and $+90^\circ$ hue rotation
Cue conflicts	AdaIN strength 0.8, five class pairs in both directions, 200 accepted images
Translation	0, 8, 16, and 32 pixels in four directions, reflection padding
Patch shuffle	One seeded permutation of a 4×4 grid per image
Representation analysis	Paired cosine similarity, t-SNE with 100 matched images per condition, cosine metric, perplexity 30, PCA initialization, and seed 6304

classification decision, while object structure and geometry remain. Hue rotation produces somewhat greater degradation (2.6–4.4 points) relative to the clean baselines. If a model associated a class with an object in a specific colored setting, changing those colors might affect its decision; further controlled interventions would be required to establish this explanation.

When we shuffle image patches, pixels and details within each patch remain preserved, but the relationships between patches change. This disrupts the image’s global spatial arrangement, making its overall shape harder to recognize, consistent with the 8–14-percentage-point drop in accuracy across the models.

Table 2: Clean performance and color and patch-shuffle results on 500 STL-10 test images. Gray, Hue, and Shuffle report accuracy / prediction consistency (%); clean accuracy and macro-F1 are percentages, and confidence is mean maximum class probability.

Model	Clean	Macro-F1	Confidence	Gray	Hue $+90^\circ$	Shuffle
ResNet-50	97.4	97.41	0.9091	95.6 / 96.2	93.0 / 94.0	88.2 / 89.8
ViT-B/16	97.4	97.39	0.9694	95.2 / 95.2	94.0 / 94.4	89.4 / 89.8
CLIP head	98.4	98.40	0.2150	96.4 / 96.6	95.8 / 96.4	85.0 / 85.6
CLIP zero-shot	98.0	97.99	0.9548	97.0 / 97.2	95.2 / 96.0	84.0 / 84.0

In investigating the extent to which the models rely on shape or texture, we consider the cue-conflict results in Table 3. All models select the shape label more often than the texture label, giving shape-bias scores of 86.93–94.74% with coverage of 88–95%. These results suggest a preference for shape on the accepted images, but the conclusion depends on how clearly AdaIN transferred the intended texture. Some stylizations appeared noisy or conveyed color and background more clearly than texture. The visual rejection rule also involves judgment, and our STL-10 generation procedure differs from that in Geirhos et al. (2019), *ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness*, so numerical bias scores are not directly comparable.



Figure 1: Selected cue-conflict examples and model predictions.

The airplane example in Figure 1 was borderline accepted under our visual review rule. Its shape remains clearly recognizable after stylization, but the intended texture is difficult to identify. Much of the transferred appearance resembles noise. We observed this problem in many generated candidates. Consequently, a model selecting the airplane label in this example does not provide strong evidence that it preferred shape over a recognizable competing texture. This limitation may contribute to the high shape-bias scores in Table 3.

We therefore ran a separate, controlled style-transfer experiment with a foreground mask and repeated texture crop. Shape bias fell to 71.70–76.92%, while coverage fell to 53–76%. More predictions selected the texture label, but many more selected neither nominated class. This shows why shape bias and coverage must be interpreted together: high coverage does not establish that the texture cue was valid, while lower coverage makes a shape-bias percentage less representative of all predictions. The procedure and selected examples are presented in the Appendix.

Table 3: Predictions on 200 accepted shape–texture cue conflicts. Of 440 generated candidates, 200 were accepted, 111 rejected, and 129 unreviewed.

Model	Shape	Texture	Other	Shape bias (%)	Texture bias (%)	Coverage (%)
ResNet-50	153	23	24	86.93	13.07	88
ViT-B/16	171	13	16	92.93	7.07	92
CLIP head	180	10	10	94.74	5.26	95
CLIP zero-shot	178	10	12	94.68	5.32	94

Figure 2 shows that shifting an image by 16 pixels causes little change in top-1 accuracy or prediction consistency across all four models. Accuracy remains between 96.75% and 97.90%, while consistency remains between 98.30% and 98.90%. A convolution can detect a local pattern at a shifted position, which helps explain why ResNet’s predictions remain stable, although the full network is not exactly translation invariant. ViT uses positional embeddings, yet it also remains stable under this shift; thus, the results do not support a simple division between CNNs and transformers. CLIP’s trained head and zero-shot classifier use the same image encoder but different decision rules, and both remain stable. Their architecture and training differences cannot be isolated from this comparison alone.

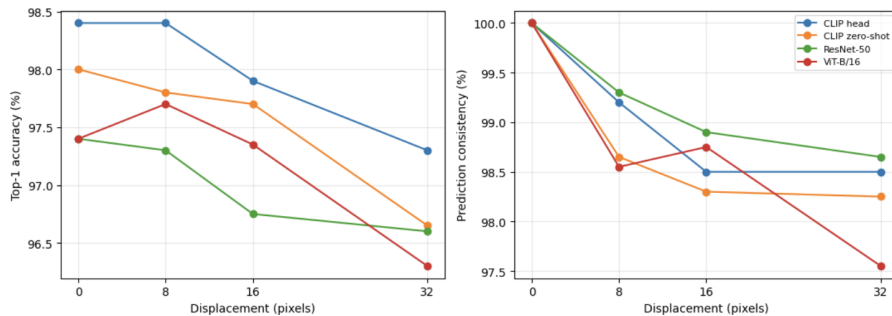


Figure 2: Accuracy and prediction consistency as a function of translation distance, averaged over four directions.

Table 4 compares each transformed image’s representation with its clean counterpart. Translation by 16 pixels produces the highest cosine similarity for all three backbones (0.9525–0.9743), consistent with stable predictions in Figure 2. Grayscale has a smaller effect on accuracy, yet its feature similarity is only 0.7393 for ViT and 0.7885 for ResNet. Thus, a model can retain its predicted class even when its representation changes appreciably. Cue conflicts and patch shuffling produce larger feature changes, although their cosine values alone do not identify which visual cue controls the final decision.

Figure 3 provides a qualitative view of the feature spaces. Clean and translated examples generally remain in similar class regions. Patch-shuffled examples show more mixing across classes, particularly for CLIP, where many shuffled points move toward the central region of the plot. Cue-conflict examples also move away from clean neighborhoods in some cases, but their positions alone cannot establish whether shape or texture controlled a prediction; the decision counts in Table 3 address that question. Since t-SNE does not preserve all feature-space distances, we use these plots to inspect neighborhoods rather than to quantify representation change.

Table 4: Mean cosine similarity between clean and transformed features.

Backbone	Grayscale	Cue conflict	Patch shuffle	Translation (16 px)
ResNet-50	0.7885	0.5442	0.7150	0.9525
ViT-B/16	0.7393	0.5772	0.6538	0.9743
CLIP ViT-B/32	0.8983	0.7934	0.7305	0.9680

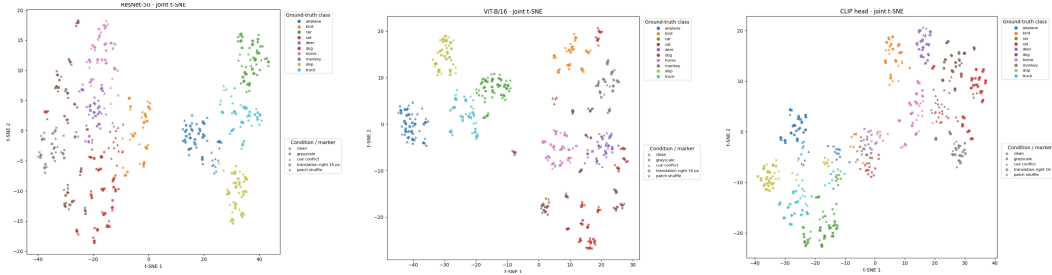


Figure 3: Joint t-SNE projections of clean and transformed feature representations for ResNet-50, ViT-B/16, and CLIP, respectively.

Taken together, these experiments allow us to revisit the research questions and summarize what the observed changes in predictions and representations support.

1. **Color and cue conflict.** Grayscale causes little degradation in accuracy or prediction consistency, while shape labels dominate the accepted cue-conflict images. Together, these results are consistent with models retaining their decisions when color changes or the intended texture conflicts with shape. They do not establish that shape is the controlling cue: AdaIN stylization quality varies, and the selected class pairs, sample size, and visual rejection rule affect the comparison. In the controlled style-transfer run, more predictions select the texture label, but coverage also falls because more predictions select neither nominated class. Shape bias, texture counts, and coverage must therefore be interpreted together.
2. **Translation versus patch shuffling.** The models tolerate moderate displacement but depend more strongly on relationships between image parts. Convolution can detect a local pattern after it moves, which may contribute to ResNet’s translation stability. ViT’s attention connects tokens across the image, but its positional embeddings and the changed relationships between shuffled patches mean that attention does not make its predictions invariant to shuffling. CLIP’s image encoder is also a ViT, while its trained head and zero-shot classifier apply different decision rules. Isolating the contribution of each mechanism would require further controlled interventions.

3. **Predictions versus representations.** One possible explanation for the grayscale mismatch is that color removal changes the representation without changing the features or decision margins that determine the predicted class. The cosine score alone cannot support that explanation. CLIP’s trained head and zero-shot classifier share the same image encoder but have different prediction consistencies after shuffling (85.6% and 84.0%, respectively). This shows that the classification rule also affects how shared image features produce a label.
4. **Explaining model differences.** ResNet-50 selects the texture label more often than ViT-B/16 on the accepted cue conflicts (23 versus 13 of 200), while all models remain relatively stable under translation. ResNet’s local convolutional operations provide a plausible explanation for some translation stability, but these results cannot isolate an architectural cause. The backbones also differ in pretrained weights, supervision, augmentation, and capacity. Comparing CLIP’s trained head with its zero-shot classifier more directly probes the effect of the classification rule because both use the same image encoder. We therefore interpret differences between backbones as observed behavior under these interventions, rather than as effects caused solely by architecture.

3 Unsupervised Domain Adaptation

We previously examined which visual cues the models use, how interventions affect their predictions, and whether those changes are reflected in their feature representations. We now ask whether representations and decision rules learned from labeled source domains transfer to a visually different target domain whose labels are unavailable during training. This is *unsupervised domain adaptation*: learning remains supervised on the source domains, while target examples provide no class-label supervision.

A domain is characterized by its input space and data distribution. Source and target inputs may differ in their marginal distributions, $P_S(X) \neq P_T(X)$, as in a change from photographs to sketches. Under a pure covariate shift, their labeling relationship remains the same, $P_S(Y | X) = P_T(Y | X)$. If the conditional relationship also changes, $P_S(Y | X) \neq P_T(Y | X)$, aligning the input distributions alone may fail to preserve class information.

We compare four methods. Source-only ERM trains on labeled source images without target adaptation and provides the baseline. DAN adds a Maximum Mean Discrepancy penalty to align source and unlabeled target features. DANN uses a domain discriminator and gradient reversal to encourage domain confusion. CDAN conditions that discriminator on both image features and predicted class probabilities, aiming to align domains while retaining class information. We also vary DAN’s alignment strength to examine the resulting trade-offs.

This section addresses four research questions:

- **RQ1:** How large is the Source-only source-to-target gap, and which classes contribute most to its failures?
- **RQ2:** Does lower domain separability correspond to better target recognition across Source-only, DAN, DANN, and CDAN?
- **RQ3:** Does CDAN’s class-conditional alignment preserve class structure better than the marginal alignment used by DAN and DANN?
- **RQ4:** How does alignment strength affect source performance, domain separability, and target performance, and which setting could be selected without target labels?

3.1 Experimental Design and Setup

We use PACS with Photo, Art Painting, and Cartoon as labeled sources and Sketch as the unlabeled target. Source-only ERM establishes performance under supervised learning using source label distribution without any adaptation. DAN tests whether matching source and target feature distributions using MMD improves transfer by learning domain invariant features that are causally related to labels for the target domain. DANN tests adversarial domain confusion, while CDAN also supplies predicted class probabilities to the discriminator to test whether this helps retain class information. All methods share the protocol in Table 5; only their alignment objectives differ.

We select checkpoints using mean macro-F1 across the three source validation domains. Sketch labels are used only after settings and checkpoints have been fixed. We report source and target accuracy and macro-F1, then measure domain separability using a classifier trained on frozen source-validation and target features. Per-class accuracy changes and confusions show whether an aggregate result hides negative transfer. For the controlled study, we vary DAN’s MMD weight over $\{0.1, 1, 10\}$. We expect stronger alignment to reduce domain separability, although excessive alignment may weaken source discrimination and target recognition.

Initial training produced large feature norms and unstable gradients. We therefore clip global gradients to norm 1.0 for every method. For DANN and CDAN, we normalize the discriminator’s feature input and initialize its final layer to zero. The class head still receives the original features. These changes were selected from source-side training results before observing Sketch labels.

Table 5: Experimental Settings.

Setting	Configuration
Data	PACS; Photo, Art Painting, Cartoon sources; Sketch target; stratified source 80/20 splits; seed 6304
Backbone	ImageNet1K V1 ResNet-18, fully fine-tuned; seven-class linear head; pre-trained BatchNorm statistics frozen, affine parameters trainable
Images	Resize 256×256 ; random 224×224 crop and horizontal flip for training; center crop for evaluation; pretrained normalization
Training	Eight images per source domain and 24 unlabeled target images per update; AdamW, learning rate 10^{-4} , weight decay 10^{-4} ; maximum 30 epochs; patience 5
DAN	MMD on 512-dimensional features; three RBF bandwidths 0.5, 1, 2 times the batch median squared feature distance; main weight $\lambda = 1$
DANN/CDAN	256-unit discriminator, ReLU, dropout 0.5, two-class output; unit domain-loss weight; $\alpha(p) = 2/(1 + \exp(-10p)) - 1$
CDAN	Discriminator input $f \otimes p$, using features and predicted class probabilities without detaching either
Stabilization	Gradient clipping at 1.0 for all methods; discriminator features normalized and multiplied by $\sqrt{512}$, with zero-initialized final layer, for DANN/CDAN
Diagnostic	Equal source-validation and target feature counts; seeded 70/30 split; balanced logistic regression, $C = 1$

3.2 Results and Discussion

We first examine source-validation performance in Table 6. Source-only ERM achieves 93.29% mean accuracy and 93.01% mean macro-F1. Its highest accuracy is on Photo (95.21%), while Art Painting is lowest (90.00%), suggesting that this source domain is more difficult under the shared training procedure and settings. DAN at $\lambda = 0.1$ improves Photo accuracy to 97.60% and mean source accuracy to 95.01%. DAN at $\lambda = 1$ also retains strong source performance, whereas increasing the alignment weight to 10 reduces mean source accuracy to 21.66% and macro-F1 to 5.07%. This collapse across Photo, Art Painting, and Cartoon is consistent with the alignment penalty dominating over the supervised classification objective, although the accuracy results cannot establish this alone. DANN and CDAN retain Photo accuracies of 95.81%, close to Source-only, but their performance across all three source domains must also be considered.

We next examine performance on the Sketch target domain in Table 6. Source-only ERM achieves 60.73% accuracy, compared with 93.29% mean source-validation accuracy, giving a 32.56-percentage-point domain gap. Its domain-separability score is 99.86%, meaning that source and Sketch features remain easy to distinguish. DAN at $\lambda = 0.1$ performs best on Sketch, reaching 72.92% accuracy while reducing separability to 93.68%. At $\lambda = 1$, separability falls further to 88.74%, but Sketch accuracy is lower at 64.85%, although still above Source-only. Increasing λ to 10 reduces separability to 61.13% while accuracy collapses to 4.07%. This suggests a trade-off: stronger alignment makes domains harder to distinguish, but excessive pressure may also weaken class-discriminative information. DANN reaches 52.74% Sketch accuracy, while CDAN falls to 15.35%. Neither improves on Source-only in this run. Thus, lower domain separability does not by itself guarantee better target classification.

Table 6: Source-validation and Sketch target results on PACS. Accuracies, macro-F1, and domain separability are percentages; Δ is the Sketch accuracy change relative to Source-only in percentage points. Each source entry reports accuracy / macro-F1.

Source validation					
Method	Epoch	Photo	Art Painting	Cartoon	Mean
Source-only	3	95.21 / 94.54	90.00 / 89.59	94.67 / 94.92	93.29 / 93.01
DAN 0.1	7	97.60 / 97.12	91.46 / 91.50	95.95 / 96.42	95.01 / 95.01
DAN 1	3	95.81 / 95.23	91.46 / 91.23	94.88 / 95.52	94.05 / 94.00
DAN 10	1	25.75 / 5.85	21.95 / 5.14	17.27 / 4.21	21.66 / 5.07
DANN	7	95.81 / 94.91	89.76 / 89.81	94.67 / 94.95	93.41 / 93.22
CDAN	8	95.81 / 95.16	87.56 / 87.98	93.82 / 94.25	92.40 / 92.46

Sketch target and domain diagnostic				
Method	Target Acc.	Target F1	Δ Acc. (pp)	Separability
Source-only	60.73	57.16	0.00	99.86
DAN 0.1	72.92	67.73	+12.19	93.68
DAN 1	64.85	57.71	+4.12	88.74
DAN 10	4.07	1.12	-56.66	61.13
DANN	52.74	52.36	-7.99	99.73
CDAN	15.35	16.43	-45.38	99.86

Table 7: Source-only per-class accuracy on Sketch (%) and changes relative to Source-only for each adaptation method (percentage points).

Class	Source-only	DAN 0.1	DAN 1	DAN 10	DANN	CDAN
Dog	11.27	+29.8	+25.6	-11.3	+60.0	-11.0
Elephant	98.11	-20.0	-33.1	-98.1	-17.3	-98.1
Giraffe	63.35	+13.0	+12.1	-63.3	-43.4	-56.6
Guitar	65.95	+29.9	+18.4	-66.0	+8.6	+8.1
Horse	74.02	-2.9	-1.3	-74.0	-47.7	-72.8
House	71.25	+28.7	+28.7	-71.2	-20.0	+28.7
Person	21.25	+73.8	-3.8	+78.8	+19.4	-15.0

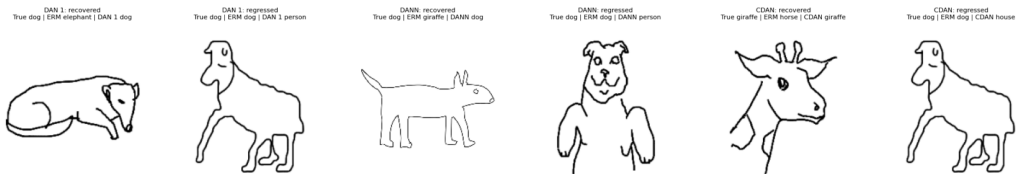


Figure 4: Selected Sketch images showing predictions recovered or lost by DAN, DANN, and CDAN relative to Source-only ERM.

Table 7 shows that overall gains can hide class-specific failures. DAN at $\lambda = 0.1$ improves person accuracy by 73.8 percentage points and guitar accuracy by 29.9 points, but reduces elephant accuracy by 20.0 points. CDAN loses 98.1 points on elephant and 72.8 points on horse. It predicts house for 733 elephants and 804 horses. These concentrated confusions help explain CDAN’s poor Sketch accuracy. Thus, class-conditioned alignment did not preserve the distinctions between these classes in this experiment, while even the best-performing DAN setting produced some negative transfer.

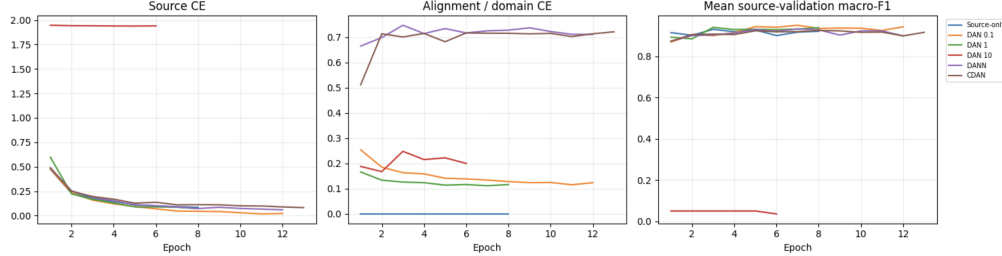


Figure 5: Training classification loss, alignment or domain loss, and mean source-validation macro-F1 across epochs for Resnet-18.

Figure 5 helps explain the performance differences. For most methods, source classification loss falls and source-validation macro-F1 remains high. DAN at $\lambda = 10$ is the only exception here. Its classification loss remains high and macro-F1 stays near 0.05. The plotted alignment values are different quantities for DAN and the adversarial methods, so their magnitudes should not be compared directly generally. The curves also show why source validation alone cannot establish successful adaptation as CDAN maintains high source-validation performance while having poor performance on the sketch target domain.

Lastly, we use our above results to revisit the four research questions:

1. **Source-to-target gap.** Source-only ERM falls from 93.29% mean source accuracy to 60.73% on Sketch, a gap of 32.56 percentage points. The class-level results show that adaptation gains and failures are generally not equal. At the class level, Source-only performs worst on dog (11.27%) and person (21.25%) Sketch images. Dog contributes particularly strongly to the overall error because it has 772 examples, of which 685 are misclassified.
2. **Domain separability.** Lower separability does not grant better Sketch recognition. DAN at $\lambda = 10$ reduces domain-probe accuracy to 61.13%, but achieves only 4.07% Sketch accuracy while having lower domain separability. DAN at $\lambda = 0.1$ achieves 72.92% Sketch accuracy despite higher probe accuracy of 93.68%.
3. **Class-conditional alignment.** CDAN’s conditioning does not improve transfer in the experiment. We find its Sketch accuracy to be 15.35%, and many elephant and horse images are classified as houses, suggesting a clear class confusion. DAN at $\lambda = 0.1$ performs best overall, although its elephant accuracy declines.
4. **Alignment strength and selection.** Increasing DAN’s weight to 10 leads to a collapse in both source and target classification, even as domain separability decreases. Among the tested settings, $\lambda = 0.1$ could have been selected by its highest mean source-validation macro-F1, without using Sketch labels.

4 Domain Generalization

We previously examined whether models that are trained in a source-only domain can adapt to a target domain. We now remove access to Sketch during training and model selection. In this setting, called *domain generalization*, the model learns from the labeled Photo, Art Painting, and Cartoon domains and is evaluated on Sketch only after all training decisions have been fixed. This lets us investigate whether learning from varied source domains is sufficient to generalize to an unseen visual domain.

We compare three approaches. ERM minimizes classification loss across the labeled sources and provides the baseline. DAN-DG adds a penalty that makes feature distributions from pairs of source domains more similar to encourage learning domain invariant features. Unlike the evaluation of DAN

previously, it does not use Sketch images here. We also explore Sharpness Aware Minimization (SAM). SAM seeks parameters whose source classification loss remains stable under a small parameter perturbation. We ask whether source-domain alignment or local stability given a loss landscape improves recognition on the unseen domain.

This section addresses four research questions:

- **RQ1:** How well do mean-source and worst-source validation performance predict Sketch performance, and which class-level failures do they hide?
- **RQ2:** Does DAN-DG reduce source-domain separability, and does this improve Sketch recognition without weakening class information?
- **RQ3:** Does SAM reduce the measured sharpness relative to ERM and DAN-DG, and does this agree with their Sketch performance?
- **RQ4:** What does the comparison between target-aware DAN and target-free DAN-DG suggest about access to unlabeled Sketch images?

4.1 Experimental Design and Setup

We reuse the PACS source splits, preprocessing, ResNet-18 initialization, training budget, and Source-only ERM checkpoint from our domain adaptation analysis. This gives us a shared baseline without retraining ERM. Each training batch contains eight labeled images from each source domain. Sketch images and labels are excluded from training, diagnostics, checkpoint selection, and hyperparameter selection. We select checkpoints using mean macro-F1 across the three source-validation domains. Sketch is loaded only for final evaluation.

For DAN-DG, we apply MMD to the 512-dimensional feature before the classifier head and average it across the Photo–Art Painting, Photo–Cartoon, and Art Painting–Cartoon pairs. The main comparison uses $\lambda_{DG} = 1$. We also test for 0.1 and 10 as a fixed alignment-strength study. SAM uses a perturbation radius of $\rho = 0.05$ and takes two forward/backward passes per update. We expect stronger source alignment to reduce source-domain separability, but excessive alignment may also harm in classification. We expect SAM to reduce the measured local loss increase, although this alone does not establish better Sketch performance.

Table 8: Experimental Settings

Setting	Configuration
Domains and split	PACS Photo, Art Painting, and Cartoon sources; unseen Sketch target, Domain Adaptation source splits, seed 6304.
Backbone	ImageNet-pretrained ResNet-18, seven-class head, frozen BatchNorm statistics, trainable affine parameters.
Images and batches	Resize 256×256 , train with random 224×224 crop and flip, evaluate with center crop. Eight images per source per update.
Training	AdamW, learning rate and weight decay 10^{-4} , gradient clipping 1.0, at most 30 epochs, selection by mean source macro-F1.
ERM	Reuse the unchanged Task 2 Source-only checkpoint.
DAN-DG	Pairwise source MMD on 512-dimensional features, three median-scaled RBF kernels, main $\lambda_{DG} = 1$, study $\{0.1, 1, 10\}$.
SAM	Two-pass SAM with radius $\rho = 0.05$. Balanced three-domain logistic probe (70/30 split, $C = 1$). Sharpness proxy on 32 validation images per source with perturbation radius 0.05.

4.2 Results and Discussion

We first examine source-validation performance in Table 9. The reused ERM baseline achieves 93.29% mean source accuracy, with Art Painting as its weakest source. SAM improves mean source accuracy to 95.91% and worst-source macro-F1 from 89.59% to 93.04%. DAN-DG at $\lambda_{DG} = 0.1$ also maintains strong source performance. However, at the weight of 1, mean source macro-F1 decreased to 5.07%, and the weight of 10 produces the same selected performance. This suggests that strong alignment between the source domains conflicts with classification performance in these experiments.

Table 9: Source-validation results for Task 3. Each domain and mean entry gives accuracy / macro-F1 (%); worst is the lowest source-domain macro-F1.

Method	Photo	Art Painting	Cartoon	Mean	Worst
ERM	95.21 / 94.54	90.00 / 89.59	94.67 / 94.92	93.29 / 93.01	89.59
DAN-DG 0.1	97.01 / 96.40	90.98 / 91.14	95.31 / 95.63	94.43 / 94.39	91.14
DAN-DG 1	25.75 / 5.85	21.95 / 5.14	17.27 / 4.21	21.66 / 5.07	4.21
DAN-DG 10	25.75 / 5.85	21.95 / 5.14	17.27 / 4.21	21.66 / 5.07	4.21
SAM	98.20 / 97.92	92.93 / 93.04	96.59 / 96.72	95.91 / 95.89	93.04

Table 10: Source-side diagnostics and final Sketch results. Separability, accuracy, and macro-F1 are percentages; Δ is the Sketch accuracy change from ERM in percentage points.

Method	Source sep.	Sharpness	Sketch acc.	Sketch F1	Δ
ERM	87.71	0.2648	60.73	57.16	0.00
DAN-DG 0.1	70.10	0.1889	57.50	57.35	-3.23
DAN-DG 1	36.21	0.0307	4.07	1.12	-56.66
DAN-DG 10	33.22	0.0236	4.07	1.12	-56.66
SAM	84.39	0.1997	73.71	75.69	+12.98

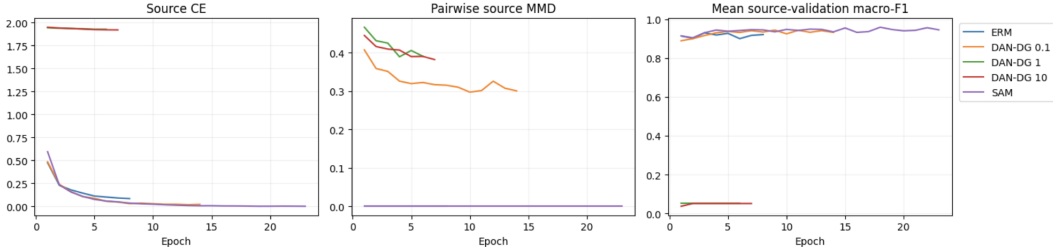


Figure 6: Source classification loss, pairwise source-domain MMD, and mean source-validation macro-F1 during training.

We next consider Sketch performance in Table 10. SAM obtains 73.71% Sketch accuracy, improving on ERM by 12.98 percentage points. DAN-DG at 0.1 performs slightly worse than ERM, despite its higher mean and worst-source scores. At weights 1 and 10, Sketch accuracy falls to 4.07%. Thus, stronger source-validation performance does not necessarily translate into better performance on an unseen domain.

Increasing DAN-DG’s alignment weight also reduces source-domain separability from 70.10% at 0.1 to 33.22% at 10, approximately the 33.3% chance level. Yet Sketch recognition becomes much worse. SAM reduces the measured sharpness increase from ERM’s 0.2648 to 0.1997 and performs better on Sketch. The collapsed DAN-DG models have still smaller sharpness values, so this result cannot be interpreted independently of their source classification performance.

Figure 6 supports this idea. Source classification loss decreases and validation macro-F1 stays high for SAM and DAN-DG 0.1. For DAN-DG 1 and 10, classification loss remains high and macro-F1 stays near 0.05. The training curves are therefore consistent with the collapse observed in both source and Sketch performance.

The class-level results in Table 11 show individual performances and failures. SAM improves dog accuracy by 56.6 points and person accuracy by 68.1 points, but reduces horse and elephant accuracy by 15.3 and 12.4 points, respectively. DAN-DG 0.1 improves dog recognition while losing 57.7 points on horse category. With DAN-DG 1, all 740 elephant images and 816 horse images are predicted as person. Selected recovered and regressed predictions are shown in Figure 7.

Lastly, we return to our research questions:

1. **Source validation and Sketch.** Mean and worst-source performance do not fully predict Sketch results: DAN-DG 0.1 improves both source measures over ERM but loses 3.23 points on Sketch. Its large horse degradation is also hidden by the overall scores.

Table 11: Per-class Sketch accuracy changes relative to ERM, in percentage points.

Class	DAN-DG 0.1	DAN-DG 1	DAN-DG 10	SAM
Dog	+29.0	-11.3	-11.3	+56.6
Elephant	-2.6	-98.1	-98.1	-12.4
Giraffe	+13.0	-63.3	-63.3	-3.6
Guitar	+5.3	-66.0	-66.0	+30.9
Horse	-57.7	-74.0	-74.0	-15.3
House	-1.3	-71.2	-71.2	+25.0
Person	+6.3	+78.8	+78.8	+68.1

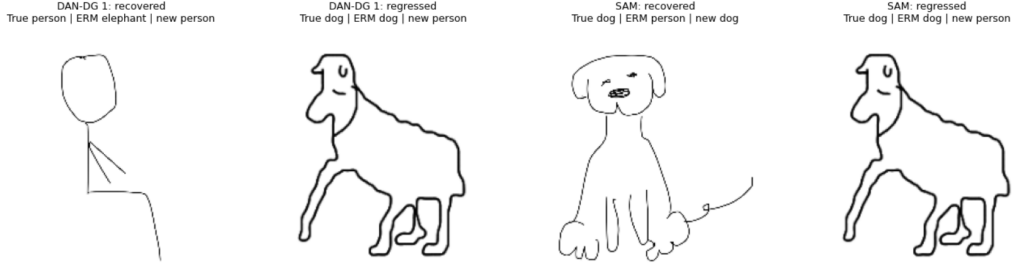


Figure 7: Selected Sketch examples showing predictions recovered or lost by DAN-DG and SAM relative to ERM.

2. **Source-domain alignment.** DAN-DG reduces separability between the observed sources, but weights 1 and 10 also collapse source and Sketch classification. Making source domains less distinguishable did not preserve the class information that was needed for this task.
3. **Local stability.** SAM improves Sketch accuracy and has a lower sharpness proxy than ERM. However, the collapsed DAN-DG models have the lowest proxy values, so the sharpness ranking does not match the Sketch accuracy ranking.
4. **Target access.** DAN at $\lambda = 1$ in the domain adaptation task achieves 64.85% Sketch accuracy, compared with 4.07% for target-free DAN-DG at the same weight. This is consistent with unlabeled target access being useful, but the two methods align different distributions and DAN-DG already fails on the labeled sources. We therefore cannot refer the entire difference solely due to target information availability.

5 Open-Set Recognition

The previous tasks evaluated models on domains that retained the same class labels. We now consider a different failure: a classifier may receive an image from a class it was never trained to recognize, but still assign it a known label. Open-set recognition (OSR) asks whether the model can retain its ability to classify known images while rejecting such unknown inputs. We use CIFAR-10 as the known label space and fixed CIFAR-100 test classes as near and far unknowns.

We first compare four post-hoc unknownness scores on the same frozen Vanilla classifier: maximum softmax probability (MSP), maximum logit score (MLS), Energy, and Mahalanobis feature distance. We then test whether stronger positive augmentation in GCSC or learned classifier and data placeholders in PROSER improve rejection. This section addresses four research questions:

- **RQ1:** Which near and far unknown classes are most often accepted, and which known labels absorb them?
- **RQ2:** How do confidence, logit magnitude, aggregate logit evidence, and feature distance differ as novelty signals?
- **RQ3:** Does GCSC improve known-class accuracy, unknown rejection, or both?
- **RQ4:** Does PROSER improve rejection beyond Vanilla and GCSC, and what happens to known-class accuracy?

Table 13: Post-hoc scores on the same frozen Vanilla model. AUROC, known-test acceptance, and unknown rejection are percentages. All thresholds accept 95% of known validation images.

Score	Near AUROC	Far AUROC	All AUROC	Known accept	Near reject	Far reject
MSP	81.53	88.95	85.24	94.74	30.88	46.38
MLS	79.65	88.94	84.30	94.78	31.88	55.75
Energy	79.65	89.05	84.35	94.77	32.12	57.75
Mahalanobis	79.99	91.19	85.59	94.60	28.38	50.75

5.1 Experimental Design and Setup

We partition the CIFAR-10 training set into 45,000 training and 5,000 validation images using seed 6304, and use its complete 10,000-image test set for final known-class evaluation. The fixed CIFAR-100 test groups contain 800 near and 800 far unknown images. No CIFAR-100 image is used for training, checkpoint selection, score construction, or threshold selection. Table 12 summarizes the experimental settings.

Table 12: Experimental Settings

Setting	Configuration
Known data	CIFAR-10, stratified 45,000/5,000 train/validation split, complete 10,000-image test set, seed 6304.
Unknown data	CIFAR-100 test set only, eight fixed near and eight fixed far classes, 100 images per class.
Backbone	Randomly initialized CIFAR ResNet-18, 3×3 , stride-1 first convolution and no initial max pool operations, 32×32 inputs.
Vanilla and GCSC	SGD, learning rate 0.1, momentum 0.9, weight decay 5×10^{-4} , cosine decay, batch size 128, 100 epochs. GCSC additionally uses RandAugment with two operations and magnitude 9.
PROSER	Initialize from selected Vanilla, append five dummy classifiers, fine-tune for 50 epochs with learning rate 10^{-3} . Classifier-placeholder weight $\beta = 1$, layer2 manifold mixup with $\lambda \sim \text{Beta}(2, 2)$ and data-placeholder weight $\gamma = 0.1$.
Selection	Highest CIFAR-10 validation accuracy, selected epochs 98 (Vanilla), 93 (GCSC), and 1 (PROSER).
Rejection	Larger scores mean more unknown. Accept an image if its score is at most the 95th percentile of the corresponding CIFAR-10 validation scores.

Vanilla and GCSC differ only in their training. PROSER uses known-class images to train dummy classifiers and mixes representations from different known classes as proxy unknowns. It does not train on real unknown images. For the post-hoc comparison, all four scores use the same saved Vanilla logits and features. We report closed-set accuracy (CSA) before rejection, AUROC for known versus near, far, and all unknowns, and rejection rates at the fixed validation-calibrated threshold. We also evaluate PROSER using both MLS on its ten known logits and its placeholder-based detection score.

5.2 Results and Discussion

We first compare scores on the frozen Vanilla model in Table 13. Its CIFAR-10 test CSA is 94.75%. All scores distinguish far unknowns better than near unknowns. MSP has the highest near AUROC at 81.53%, while Mahalanobis has the highest far AUROC at 91.19%. However, at the fixed threshold, Mahalanobis rejects fewer far unknowns than MLS or Energy.

The scores capture related but different information. MSP measures confidence relative to the other class logits. MLS retains the magnitude of the largest logit. Energy uses all logits, and Mahalanobis measures distance from known-class feature clusters. MLS and Energy rank the unknown images very similarly ($\rho = 0.996$), while MLS and Mahalanobis agree less closely ($\rho = 0.850$). For example, some wolf and fox images receive a relatively confident known-class prediction even though their logit magnitude gives different evidence of novelty. The overlapping known, near, and far score distributions in Figure 8 show why no single threshold separates all three groups.

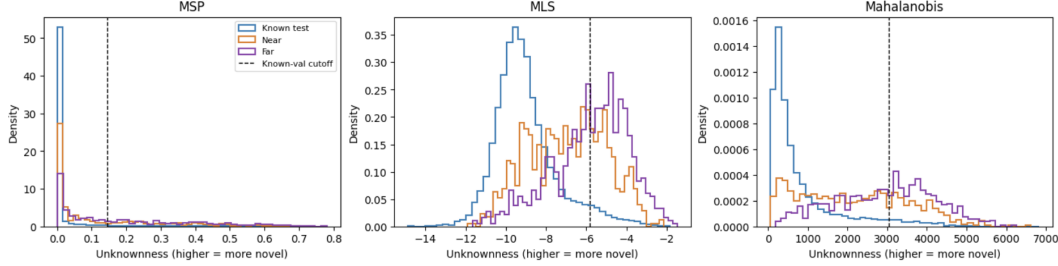


Figure 8: Unknownness-score distributions for known CIFAR-10 test images and the fixed near and far CIFAR-100 unknowns, using the frozen Vanilla model. Dashed lines mark thresholds on CIFAR-10 validation images.

Table 14: Known-class accuracy and OSR performance for trained models. CSA uses only the ten known-class logits, other entries are percentages.

Model	Score	CSA	Near AUROC	Far AUROC	All AUROC	Known accept	Near reject	Far reject
Vanilla	MLS	94.75	79.65	88.94	84.30	94.78	31.88	55.75
GCSC	MLS	94.97	80.54	91.62	86.08	94.84	32.00	60.12
PROSER	MLS	94.40	78.38	87.08	82.73	94.81	30.88	50.88
PROSER	Placeholder	94.40	76.90	88.76	82.83	94.86	30.62	54.75

Table 15: Accepted unknowns under Vanilla MLS. An image is accepted when its unknownness is at most the fixed validation threshold $\tau = -5.820$.

Group	Unknown class	Predicted known class	MLS unknownness	τ
Near	pickup truck	automobile	-11.931	-5.820
Near	bus	truck	-11.254	-5.820
Near	camel	deer	-10.756	-5.820
Far	bottle	cat	-11.664	-5.820
Far	bowl	cat	-11.235	-5.820
Far	keyboard	cat	-11.048	-5.820

We next compare trained models under the common MLS score in Table 14. GCSC raises CSA from 94.75% to 94.97%, a gain of 0.22 points. Its near rejection changes little, from 31.88% to 32.00%, while far rejection rises to 60.12%. Thus, stronger positive augmentation helps more clearly with the far unknowns in this run. PROSER instead reduces CSA to 94.40% and performs worse than Vanilla on both near and far rejection under MLS. Its placeholder score improves far rejection relative to its own MLS score, but does not exceed Vanilla or GCSC. The selected PROSER checkpoint is from its first fine-tuning epoch and later epochs had lower known-validation accuracy.

Finally, we analyze unknown images incorrectly accepted under the frozen Vanilla MLS threshold. Pickup trucks and buses are accepted most often among near unknowns, at 90/100 and 89/100 images. Their most frequent predictions are automobile and truck. Among far unknowns, bowls and mushrooms are accepted most often, at 53/100 and 50/100 images. Table 15 gives three examples from each group. The near errors have plausible visual or semantic overlap with known classes, whereas accepting a bottle, bowl, or keyboard as cat is more surprising.

In addition, we revisit the four research questions:

1. **Semantic similarity.** Near unknowns are rejected less often than far unknowns. Pickup trucks are commonly classified as automobiles and buses as trucks. Far errors such as bowls and mushrooms also remain, showing that semantic distance does not guarantee rejection.
2. **Novelty scores.** MSP, MLS, Energy, and Mahalanobis use different information and do not always rank unknowns in the same order. Mahalanobis has the strongest far AUROC, while Energy rejects the most far unknowns at the validation-calibrated operating point.

3. **Positive augmentation.** GCSC makes a small gain in known-class accuracy and improves far-unknown recognition. Its near-unknown rejection barely changes, consistent with close semantic overlap remaining difficult despite stronger augmentation.
4. **Placeholders.** The selected PROSER model does not improve rejection over Vanilla or GCSC and has slightly lower CSA. Mixing known-class representations supplies proxy unknowns between observed classes, but may not cover real unknowns arriving elsewhere in feature space. Its early selected checkpoint and validation-performance decline makes it limited about what we can conclude about the method itself.

6 Conclusion

Across the four tasks, we investigated how models use visual cues, how their predictions and representations respond to image perturbations, and whether learned information transfers across domains. Task 1 showed that preserving object structure often maintained recognition, while disrupting the arrangement of image parts caused larger losses. Task 2 asked whether unlabeled Sketch images could help align source and target representations. Task 3 removed access to Sketch and tested generalization from the observed sources alone. Task 4 then asked when a model should reject an image instead of forcing it into a known class.

Together, the results suggest that a useful representation must preserve information that distinguishes classes while becoming less sensitive to changes that do not alter the class. Invariance alone is insufficient: stronger DAN alignment made domains less separable but, when classification performance collapsed, did not improve Sketch recognition. SAM improved unseen-domain performance without explicitly aligning domains, while the OSR results showed that strong known-class accuracy still leaves many unknown images accepted as familiar classes. Robust recognition therefore depends on the visual information retained by the backbone, how the decision rule uses that information, and whether the system can recognize when its known labels do not adequately describe an input. Our experiments reveal these relationships under the tested settings, but do not establish one visual cue or training objective as universally sufficient for robustness.

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A Additional Experimental Details

A.1 Controlled Cue-Conflict Experiment

We ran a separate cue-conflict experiment to test how the stylization procedure affected the shape-bias result. Unlike the initial AdaIN run, this experiment estimated an object foreground mask using GrabCut, placed the object on a neutral background, and repeatedly tiled a central crop from the style image. It then optimized the image for 70 steps using VGG-19 content and Gram-matrix style losses, following iterative style transfer (?). The style source and content image both came from the selected STL-10 subset.

We retained the same five class pairs and both directions. A candidate was accepted only if its object remained recognizable and class-specific surface or part information from the style source was visibly transferred onto the foreground. Incorrect masks, prominent background clutter, ambiguous textures, and severe artifacts were rejected before examining model predictions. Of 960 generated candidates, 200 were accepted, 589 rejected, and 171 not reviewed. The final evaluation set contains 20 accepted images per direction.

Table 16: Controlled cue-conflict decisions on 200 accepted images per model. Shape bias is computed among shape- or texture-label predictions; coverage is their fraction among all predictions.

Model	Shape	Texture	Other	Shape bias (%)	Coverage (%)
ResNet-50	76	30	94	71.70	53.0
ViT-B/16	90	27	83	76.92	58.5
CLIP head	94	30	76	75.81	62.0
CLIP zero-shot	112	40	48	73.68	76.0



Figure 9: Examples from the controlled cue-conflict experiment, including the content image, style source, and stylized result.

A.2 Accepted Unknown Examples



Figure 10: Three near and three far CIFAR-100 unknowns incorrectly accepted under the frozen Vanilla MLS threshold. Each panel identifies the true unknown class, predicted CIFAR-10 class, unknownness score, and threshold.